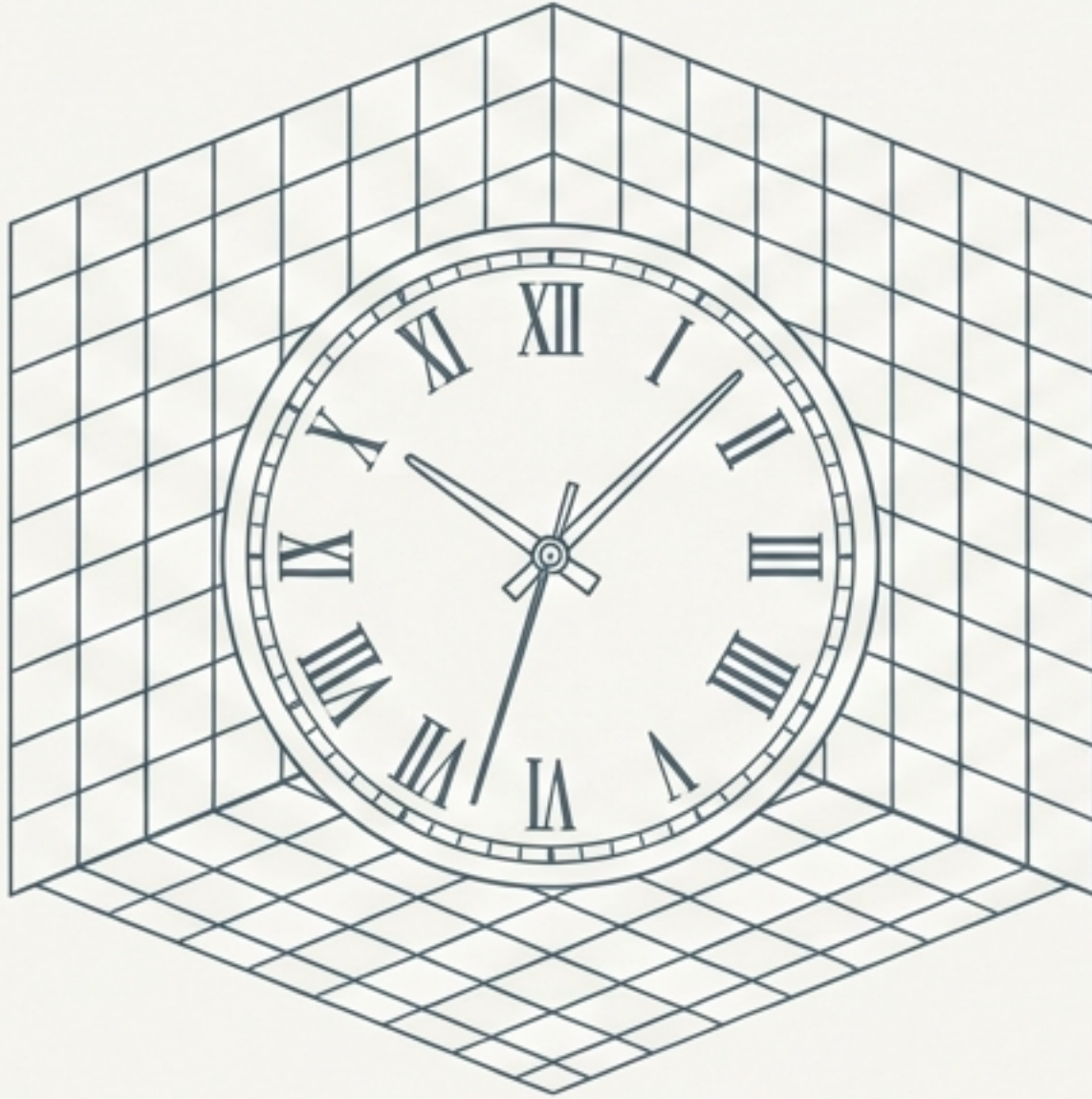


Relational Time-of-Arrival: Reading Time from the Quantum Field Itself

From Event Currents to a Continuum Clock Limit

Omar Iskandarani, Independent Researcher

In Quantum Theory, Spacetime is Dynamic. Time is Not.

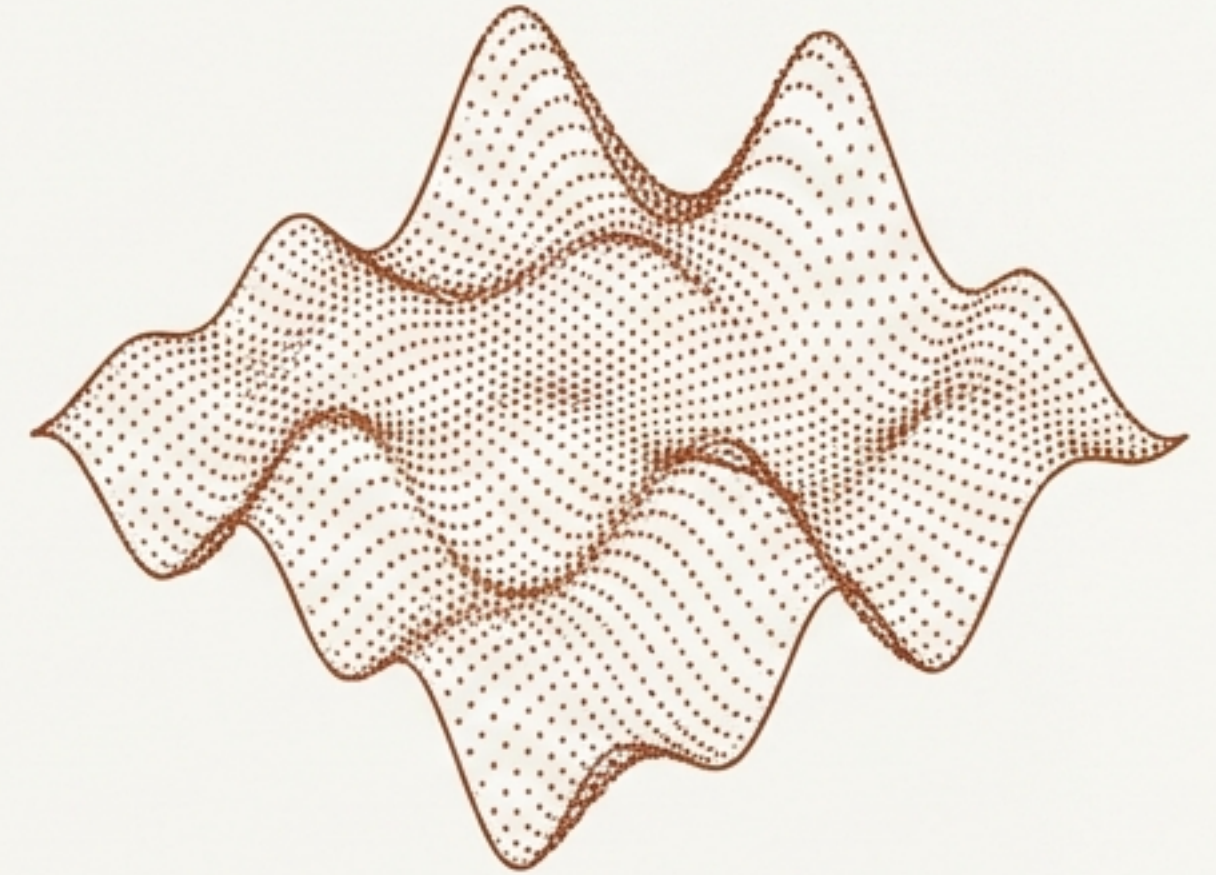


External Parameter
Time (t)

Time is usually supplied to quantum theory as an external parameter.

Operationally, however, time is read from physical degrees of freedom.

This mismatch creates fundamental problems.



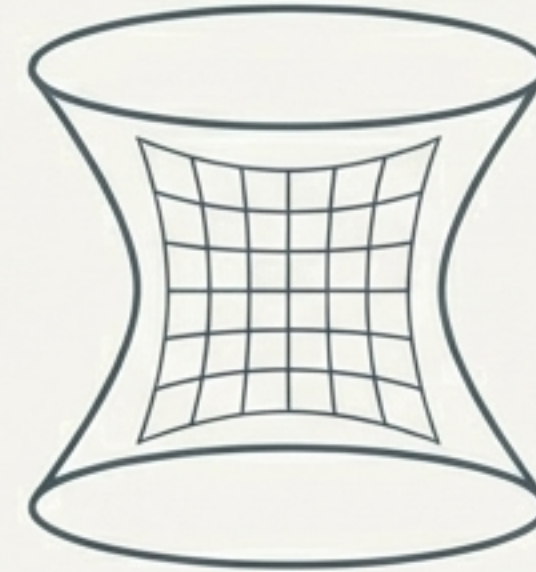
Physical Degrees of Freedom
(ψ , A_μ , etc.)

Two Fundamental Obstructions to a ‘Time Operator’



Pauli's Argument

A self-adjoint time operator, canonically conjugate to the Hamiltonian, is incompatible with Hamiltonians that are bounded from below (i.e., having a ground state).



Relativity & Localization

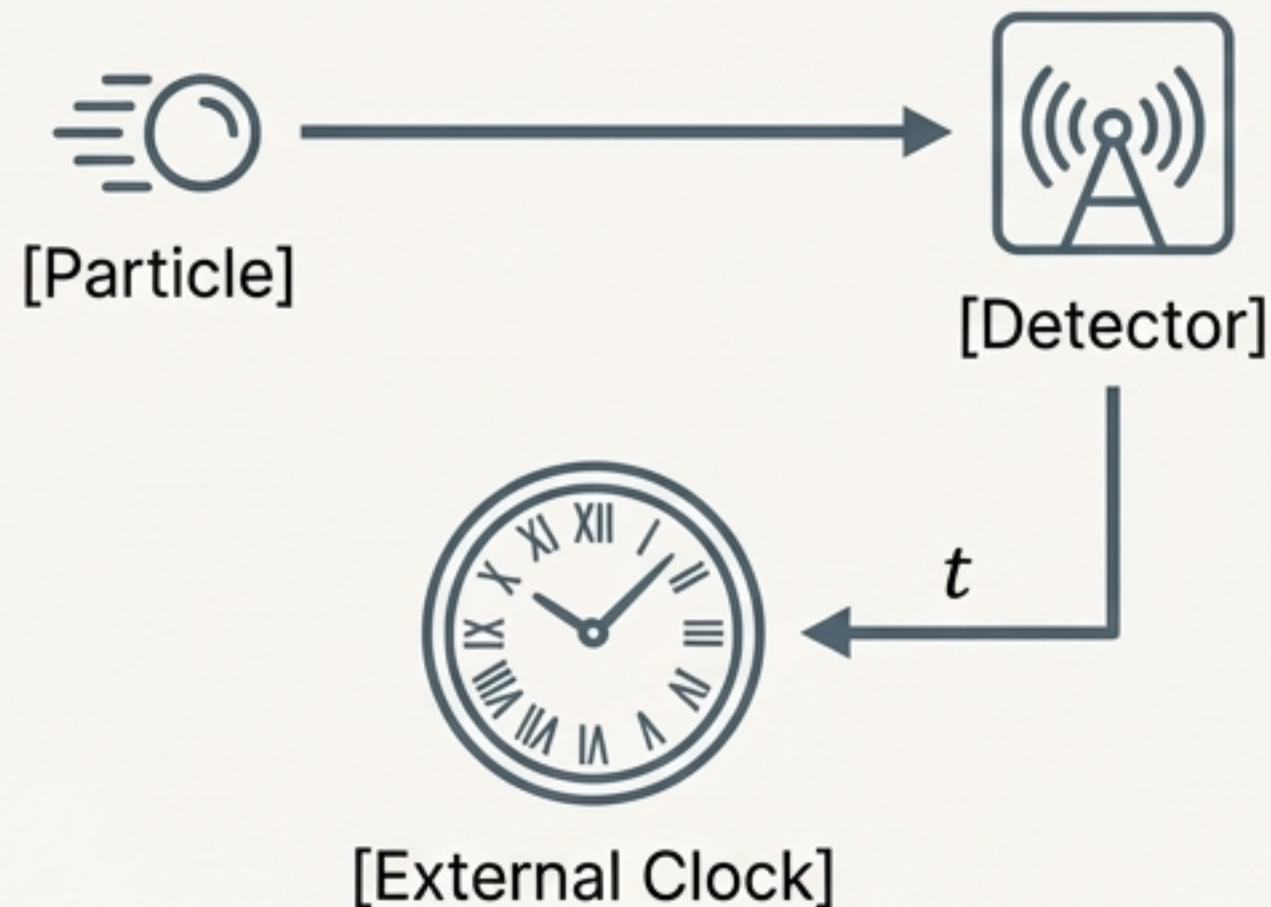
In relativistic QFT, a single, universal “time of arrival” is ill-defined. Localization is frame-dependent, and any definition must contend with causality from the start.

The absence of a universal time operator is not a gap to be patched, but a cue that TOA should be formulated relationally.

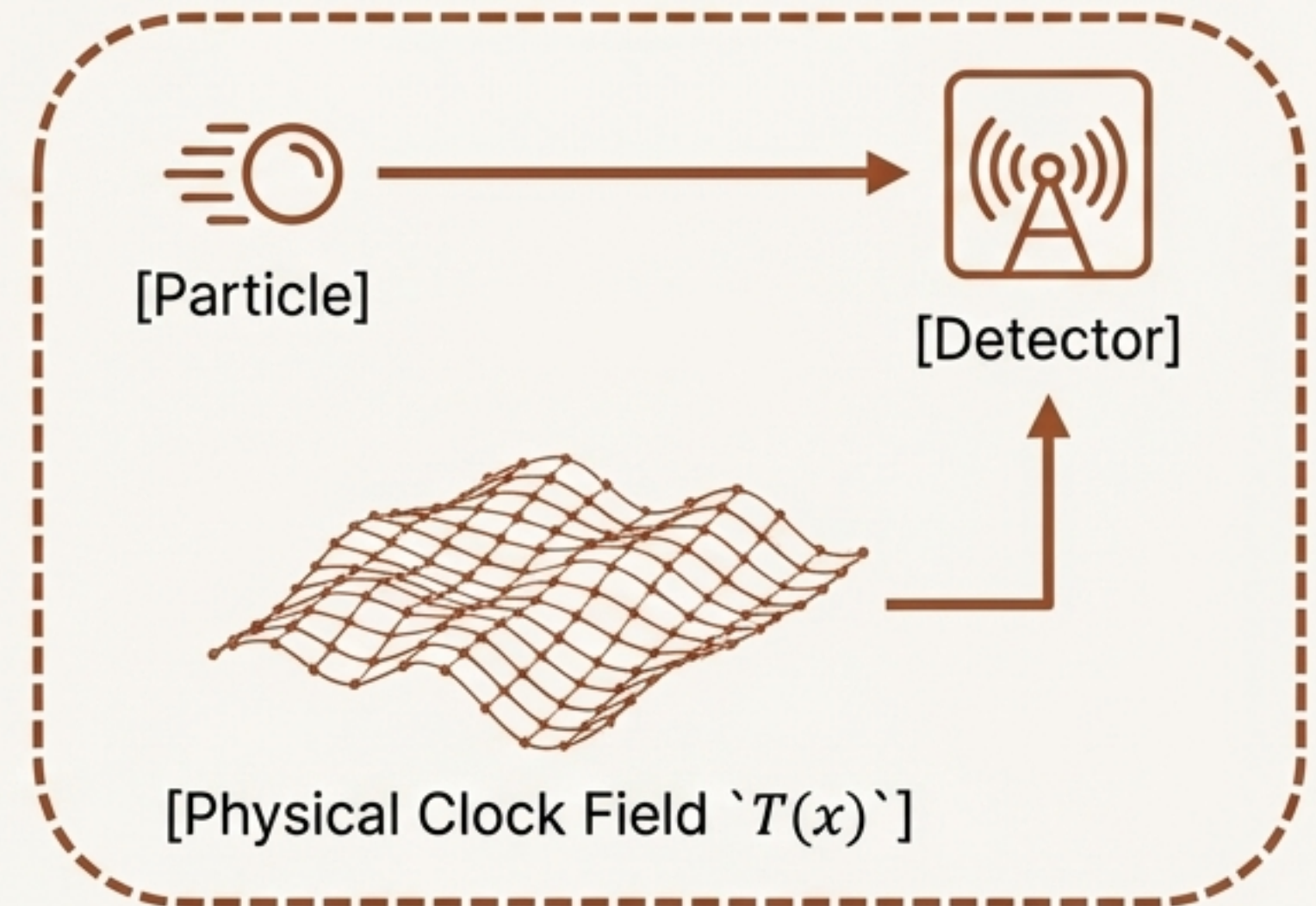
The Solution: Promote Time to a Physical Field Observable

We stop asking, “When did the particle arrive according to the universe’s clock?”
Instead, we ask, “What did the local physical clock read **when the particle arrived?**”

Old View



New View



The Foundation of the Clock: A Conserved Current of Events

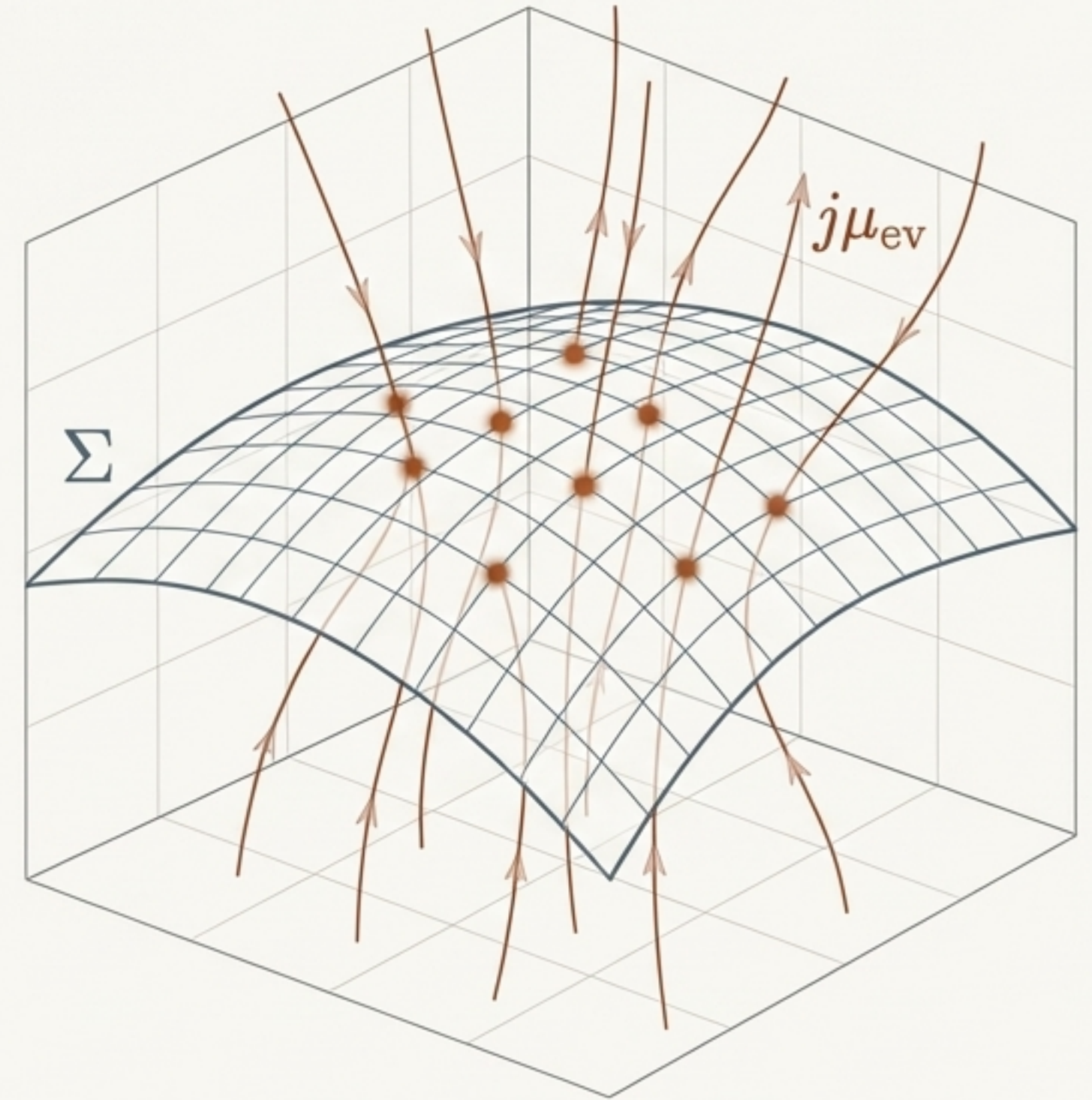
The construction begins with a conserved (or approximately conserved) **event current**, j^μ_{ev} .

$$\nabla_\mu j^\mu_{\text{ev}} \approx 0$$

This current represents the flux of localized, countable physical events.

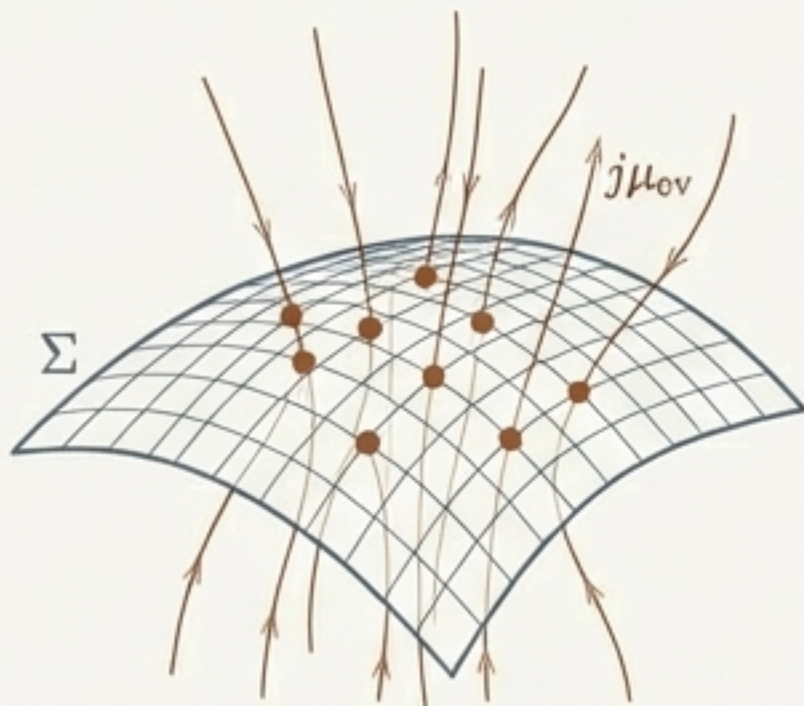
Examples: Interaction vertices, detector-registered crossings, or defect worldlines.

Conservation ensures the event count is robust under smooth deformations.



Building the Clock: From Discrete Counts to a Continuum Field

Stage 1:
UV: Discrete Events ($j\mu_{ev}$)

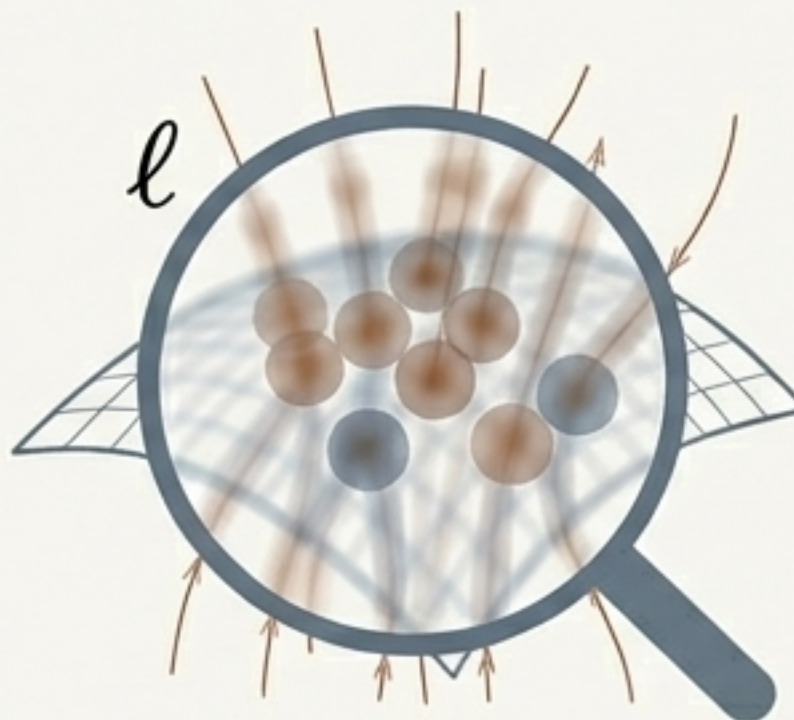


The integral of the event current over a surface Σ yields a discrete integer, the event count:

$$N_{ev}(\Sigma) \equiv \int_{\Sigma} d\Sigma_{\mu} j\mu_{ev}.$$

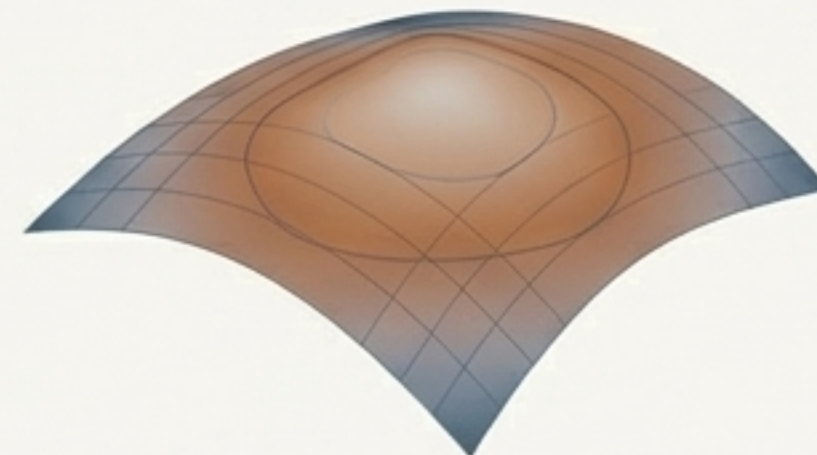
This defines the discrete clock,
 $T_{disc} = N_{ev}.$

Stage 2:
The Bridge: Coarse-Graining $(\cdot)_{\ell}$



We apply a coarse-graining operation over a spacetime scale ℓ .

Stage 3:
IR: Continuum Clock ($T(x)$)



This defines the continuum clock field $T(x)$ via

$$\partial_{\mu} T(x) \equiv \frac{1}{\nu_0} \left(j_{ev,\mu}(x) \right)_{\ell}.$$

Discrete counts and continuum time are the UV and IR descriptions of the same structure.

The Relational Time-of-Arrival Distribution

The TOA probability density with respect to the clock field T is given by:

$$p(\Theta) \equiv \frac{1}{N} \int_{\Sigma} d^3\sigma \sqrt{|h|} F(\sigma) \delta(T_{\Sigma}(\sigma) - \Theta)$$


The conserved matter flux crossing the detector world-tube Σ , defined as $F(\sigma) \equiv n_{\mu} J^{\mu}(X(\sigma))$.

The crucial constraint. This conditions the flux on the physical clock field $T(x)$ reading the value Θ at the point of arrival.

This is not a simple relabeling of a coordinate. $T(x)$ is a **dynamical field with its own correlations, which will now enter the physics.**

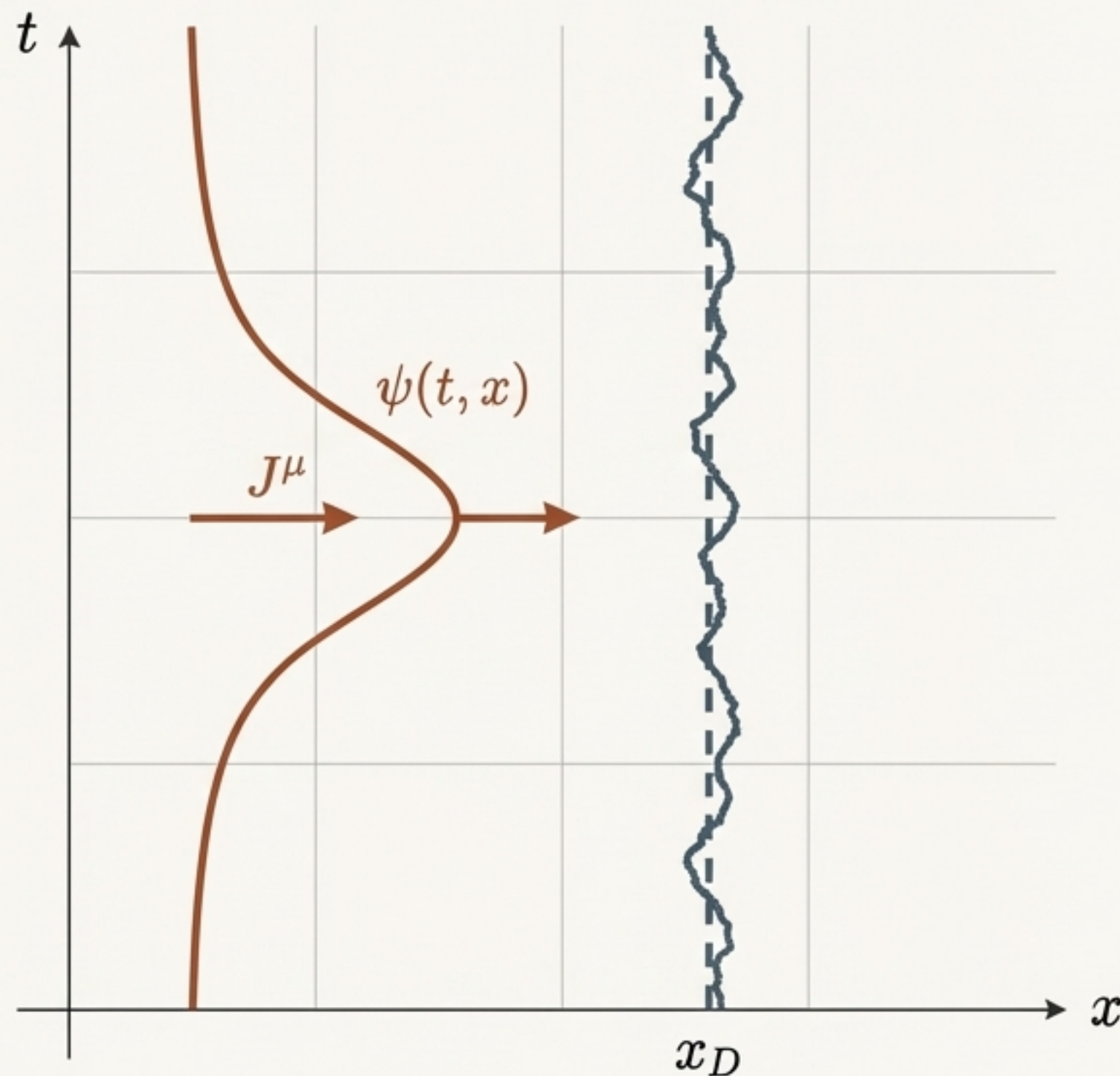
Worked Example: A Wavepacket in 1+1 Dimensions

The Setup

- **Spacetime:** 1+1D Minkowski space (t, x) .
- **Detector:** A stationary worldline at a fixed position $x = x_D$.
- **Matter:** A narrowband Klein-Gordon wavepacket $\psi(t, x)$ with a conserved current J^μ , propagating toward the detector.

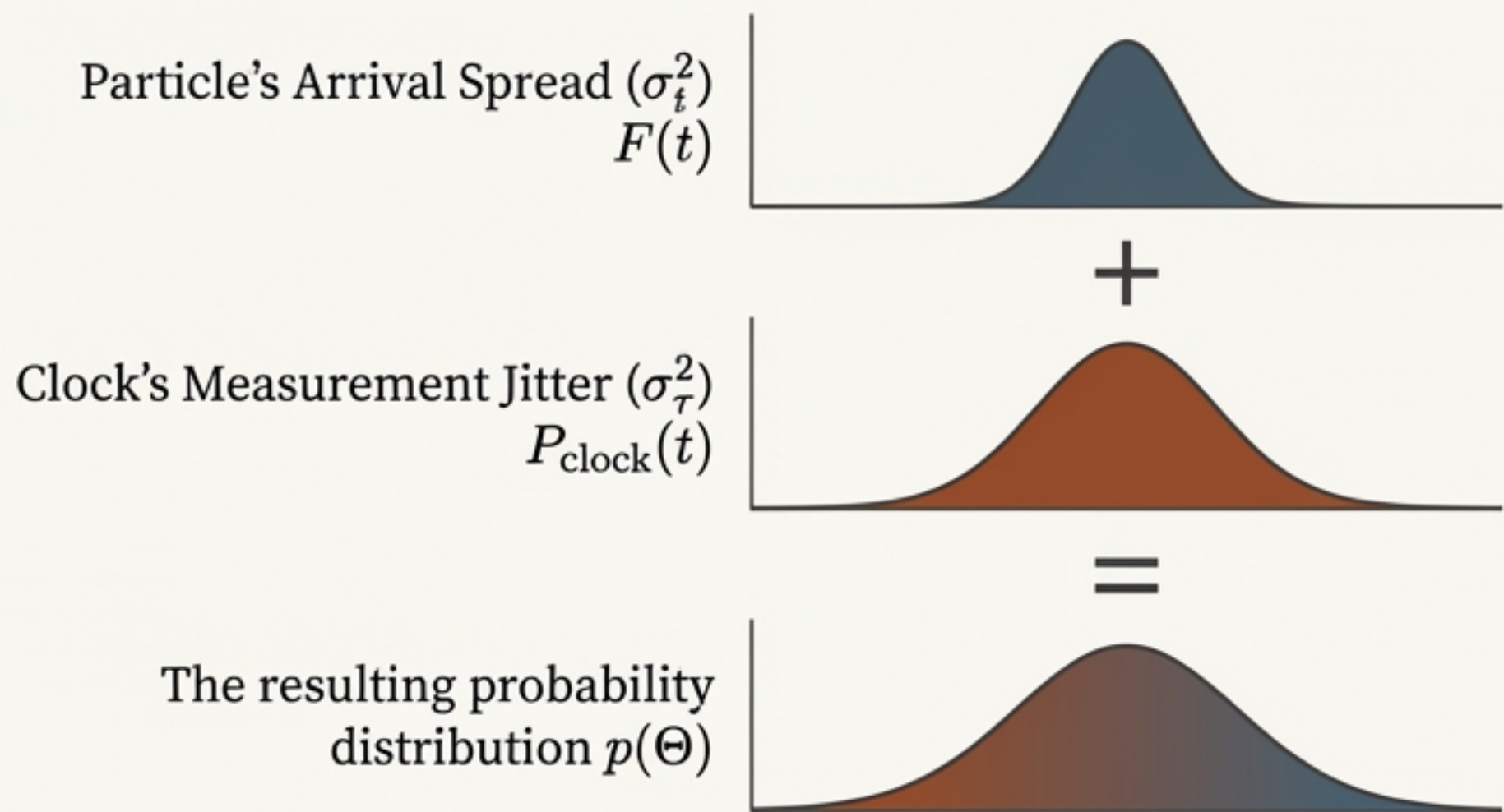
The Inputs for the Calculation

1. **Particle Flux $F(t)$:** The semiclassical flux of the wavepacket arriving at the detector. Under a narrowband approximation, this is a Gaussian profile with variance σ_t^2 .
2. **Clock Field $T(t, x)$:** Modeled as $T(t, x) = t + \tau(t, x)$, where $\tau(t, x)$ is a fluctuating field representing the clock's "jitter." We assume τ is a stationary Gaussian variable with variance σ_τ^2 .



The Result is a Convolution of Physics

$$p(\Theta) = \int dt F(t) * P_{\text{clock}}(\Theta - t)$$



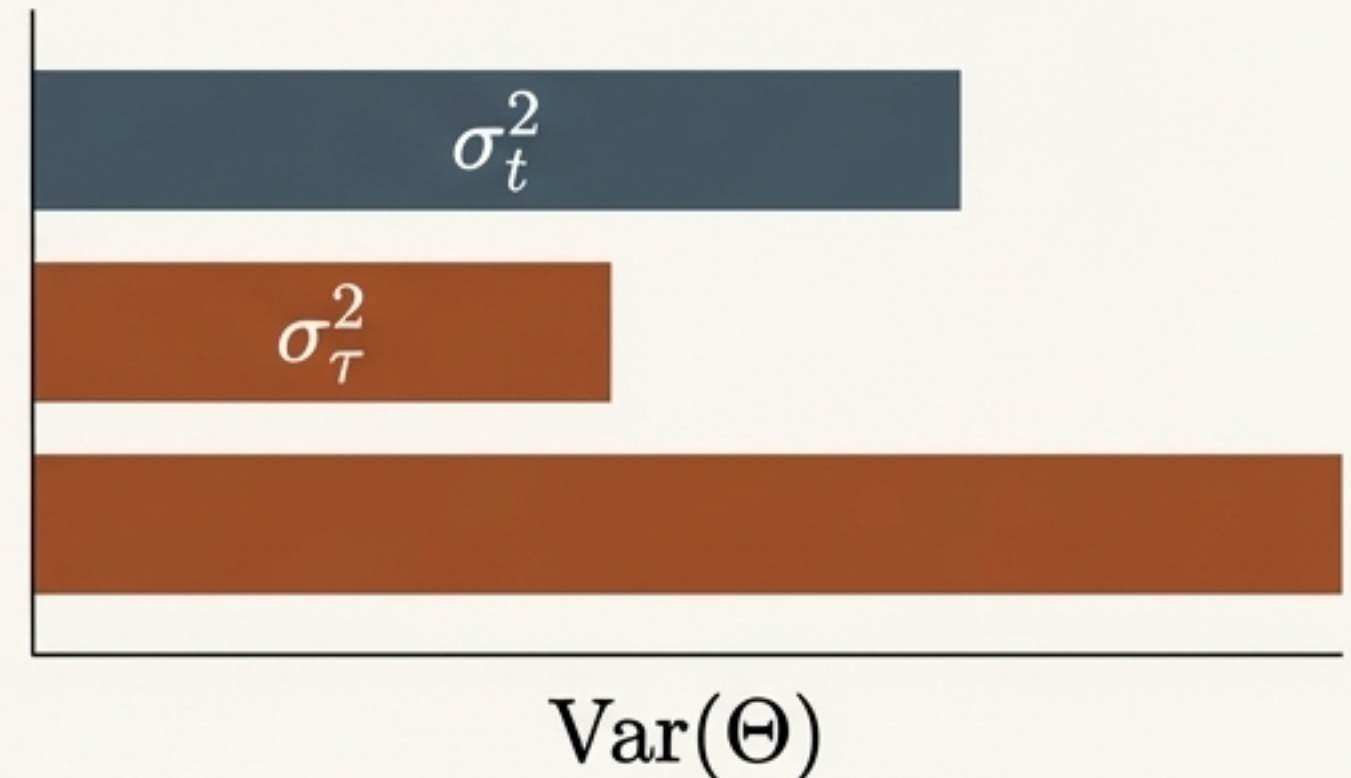
$$p(\Theta) \propto \exp \left[-\frac{(\Theta - t_{\text{cl}})^2}{2(\sigma_t^2 + \sigma_\tau^2)} \right]$$

Physical Prediction: Observable Arrival-Time Broadening

The total measured variance of the arrival time is the sum of the intrinsic spread of the matter wavepacket and the physical variance of the clock field itself.

$$\text{Var}(\Theta) = \sigma_t^2 + \sigma_\tau^2$$

- The term $\sigma_\tau^2 = \langle \tau^2 \rangle$ represents a concrete, calculable “excess” broadening of the arrival time distribution.
- This provides a direct, quantitative signature of the clock’s dynamics.
- The framework is falsifiable: deviations from this predicted variance would challenge the assumed clock sector model.



A Deeper Prediction: Massive Clocks Enforce a Causal Envelope

- The Mechanism

The clock field $T(x)$ is described by an effective field theory. A mass term μ_τ in its action sets a finite correlation scale.

- The Consequence

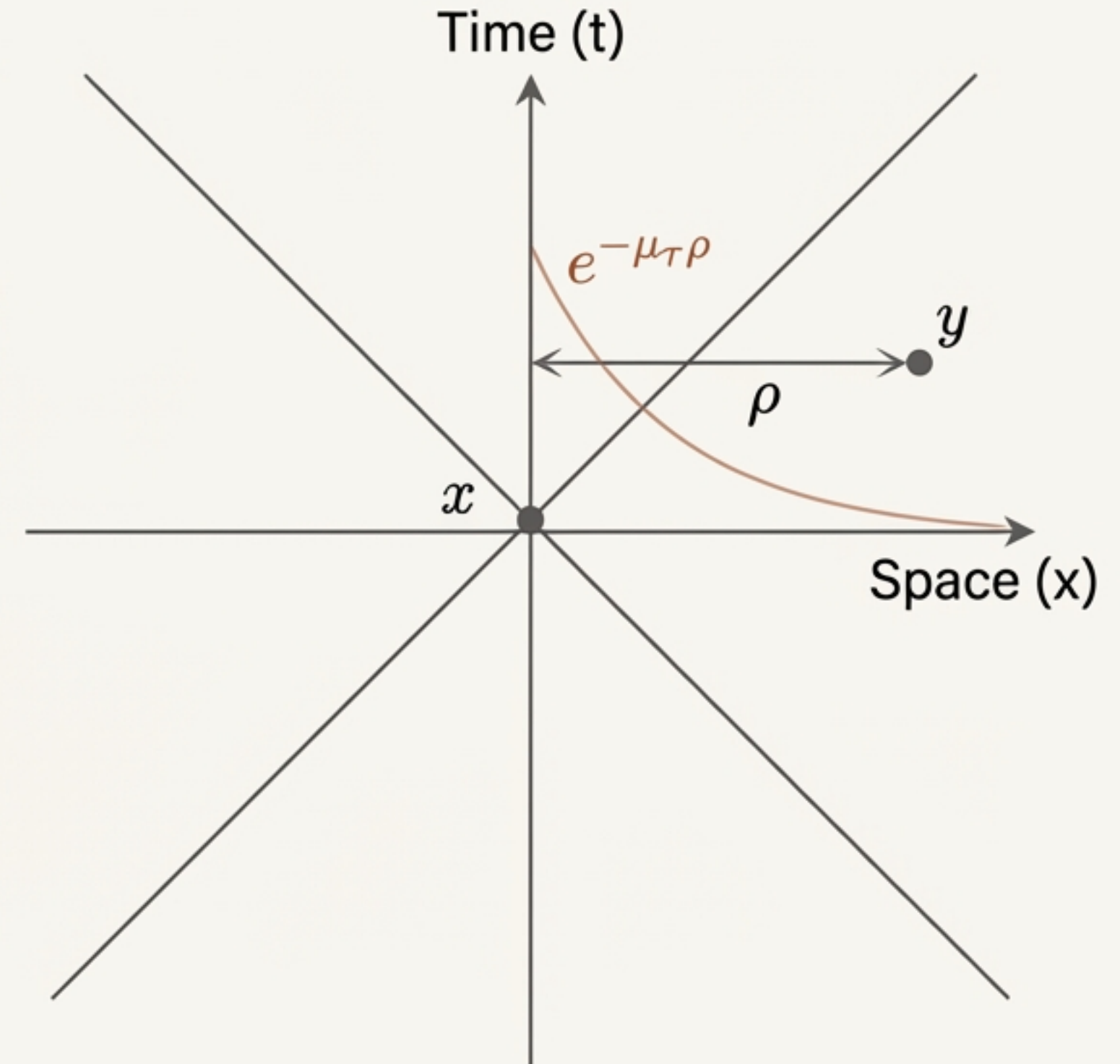
Correlations in the clock field between spacelike separated points are not strictly zero, but are exponentially suppressed.

- The Scaling Law (from Appendix A)

$$\langle \tau(x) \tau(y) \rangle \propto \frac{e^{-\mu_\tau \rho}}{\sqrt{\rho}}$$

- The Physical Insight

This provides a natural, dynamic suppression of “early” arrival probabilities that would rely on acausal correlations. The theory has a built-in mechanism to respect causality.

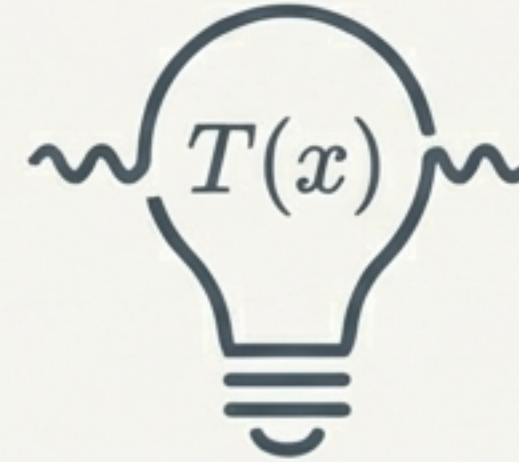


A Complete, Relational Framework for Time-of-Arrival



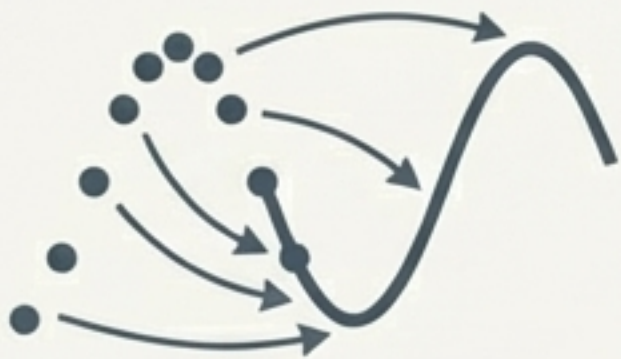
Problem

Time in QFT is an external parameter, leading to foundational obstructions (Pauli's argument, relativity).



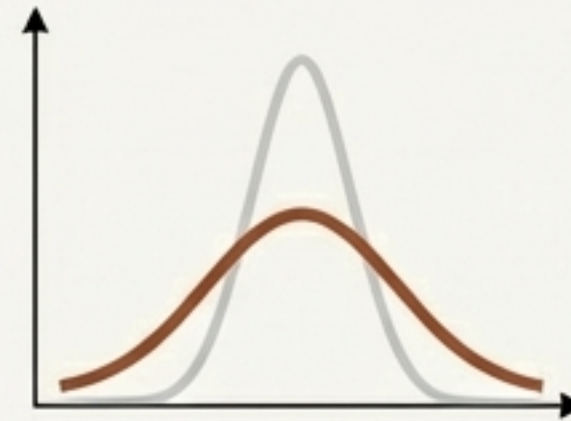
Idea

Promote time to a dynamical, relational observable by defining it via a physical clock field $T(x)$.



Mechanism

Construct the clock from a conserved "event current" (j^μ_{ev}). This provides a discrete count (N_{ev}) in the UV and a continuum field ($T(x)$) in the IR.



Predictions

The framework yields a covariant TOA distribution that predicts:

1. **Observable Broadening:**

$$Var(\Theta) = \sigma_t^2 + \sigma_\tau^2.$$

2. **Causal Envelope:** An exponential suppression $\sim e^{-\mu_\tau * \rho}$ for early arrivals.

Future Directions and Next Steps



- 3+1D Examples: Extend the calculation to 3+1 dimensions to quantify the geometric dependence of TOA on finite-size detectors.



- Non-Gaussian Clocks: Go beyond the quadratic EFT to explore non-Gaussian clock dynamics and predict their higher-cumulant signatures in the TOA distribution.



- UV/IR Crossover: Systematically study the transition from discrete, renewal-type statistics at high time resolution ($\Delta t \lesssim \ell$) to the smooth continuum limit.



- Numerical Simulations: Directly probe the predicted exponential suppression envelope by varying the clock mass μ_τ in numerical models.

Appendix: Core Definitions for Reference

Matter Current & Flux

J^μ (conserved matter current)

$F(\sigma) = n_\mu J^\mu(X(\sigma))$ (flux through detector Σ)

Event Current

$j_{\mu\text{ev}}^\mu$ (conserved event-count current)

Discrete Clock (UV)

$$N_{\text{ev}}(\Sigma) = \int_{\Sigma} d\Sigma^\mu j_{\mu\text{ev}}^\mu$$

Continuum Clock (IR)

$$\partial_\mu T(x) \equiv \frac{1}{\nu_0} (j_{\text{ev},\mu}(x))_\ell$$

Relational TOA

$$p(\Theta) = \frac{1}{N} \int_{\Sigma} F(\sigma) \delta(T(X(\sigma)) - \Theta)$$

Clock Fluctuations

$$T(x) = t + \tau(x)$$

Predicted Variance

$$\text{Var}(\Theta) = \sigma_t^2 + \sigma_\tau^2$$